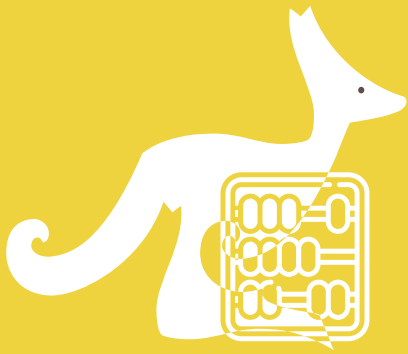




MATH KANGAROO CHINA

考试说明 / Instruction

Time allowed: 75 minutes

考试时长: 75 分钟

Language: English & Chinese

试卷语言: 英文和中文

This is a 30-question multiple-choice competition.

考试包含 30 道单项选择题。

Problems 1 to 10 are worth 3 points each.

1-10 题每题 3 分

Problems 11 to 20 are worth 4 points each.

11-20 题每题 4 分

Problems 21 to 30 are worth 5 points each.

21-30 题每题 5 分

1 point will be deducted for an incorrect answer and
no point for a blank answer.

答错扣 1 分, 不答不扣分。

The full score is 150 (Starting point is 30)

总分为 150 分 (起始分数为 30 分)

Do not open this booklet until you are told to do so.

收到监考老师指令后, 方可打开试卷册。

Mark your answers clearly on the answer sheet using a **2B pencil**.

务必使用 **2B 铅笔**在答题卡上正确填涂自己的答案。

Calculator cannot be used during the competition.

考试期间禁止使用计算器。

DO NOT discuss problems and solutions in any online or public
forum until April 30.

4 月 30 日前请勿以任何形式公开讨论试题和答案。

**Grade 9
&
Grade 10**

3 points

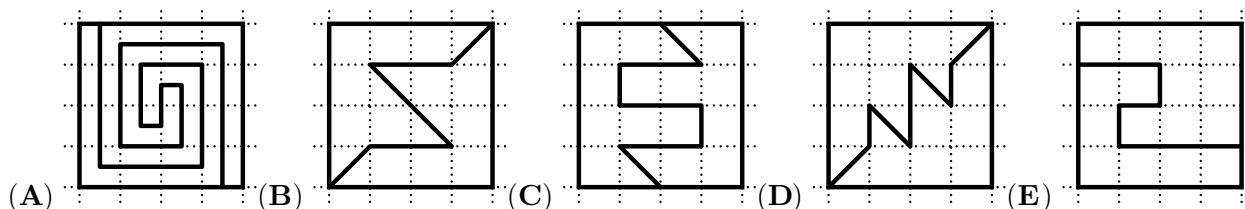
1. What is the value of $\frac{2 \times 0.24}{20 \times 2.4}$?

请问 $\frac{2 \times 0.24}{20 \times 2.4}$ 的值是多少?

- (A) 0.01 (B) 0.1 (C) 1 (D) 10 (E) 100

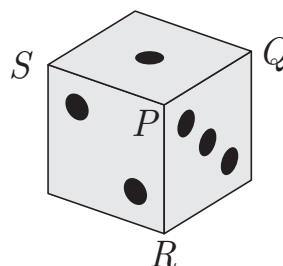
2. Which square is split up into two pieces that do **not** have the same shape?

请问下列哪个正方形被分为了两个不同的形状?



3. The number of the dots on opposite faces of a die add to 7. The vertex labelled P on the die is formed by the faces which have 1, 2 and 3 dots on them. Its vertex sum is the sum of the number of dots on those faces which meet at a given vertex. The vertex sum of P is $1 + 2 + 3 = 6$. What is the maximum of the vertex sums of vertices Q , R and S ?

一枚骰子的相对面上的点数相加等于 7。骰子上点数分别为 1、2、3 的面相交于顶点 P 。顶点之和是指相交于该顶点的所有面上的点数之和。 P 点的顶点之和为 $1 + 2 + 3 = 6$ 。请问三个顶点 Q 、 R 、 S 的顶点之和中最大值是多少?



- (A) 7 (B) 9 (C) 10 (D) 11 (E) 15

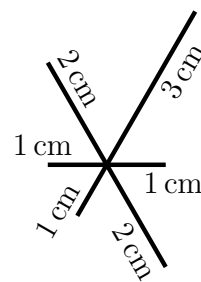
4. A hopping game is played in the following way: Each player hops into the squares, swapping between left foot - both feet - right foot - both feet - left foot - both feet, and so on, as shown. Maya played the game and hopped into exactly 48 squares starting with her left foot. How many times did her left foot touch the ground?

跳格子游戏规则为: 每位玩家跳进方格, 落脚顺序依次为左脚-双脚-右脚-双脚-左脚-双脚, 以此类推, 如图所示。Maya 参与游戏, 从左脚开始起跳, 恰好跳进 48 个格子。请问 Maya 有多少次左脚接触到了地面?

- (A) 12 (B) 24 (C) 36 (D) 40 (E) 48



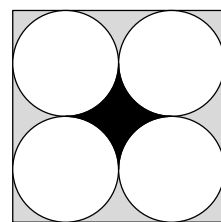
5. Tim wants to draw the figure shown on a piece of paper, without lifting his pencil off the paper. The lengths of the lines are given in the figure. He can choose to start his drawing anywhere. What is the shortest distance he could draw to complete the figure?



Tim 想在一张纸上画出右侧图形，且途中笔不能离开纸面。右侧图中已经标出各线段长度。Tim 可以选择从任意位置开始起笔作画。请问 Tim 画出这个图形所需的最短长度是多少？

- (A) 14 cm (B) 15 cm (C) 16 cm (D) 17 cm (E) 18 cm

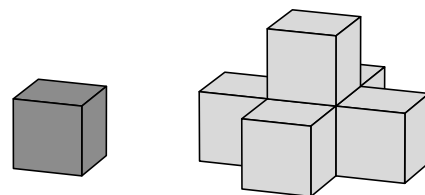
6. The figure shows a square with four circles of equal area, each touching two sides of the square and two other circles. What is the ratio between the areas of the black region and the grey region?



右侧正方形中包含 4 个面积相同的圆，其中每个圆都与正方形的两条边以及另外两个圆相切。请问黑色区域与灰色区域的面积之比为多少？

- (A) 1 : 4 (B) 1 : 3 (C) 2 : 3 (D) 3 : 4 (E) $\pi : 1$

7. John makes a sequence of structures on a table, beginning with one cube. He makes the next structure by adding five cubes which hide the visible faces of the initial cube, as shown. What is the smallest number of cubes he needs to add to the second structure so that all the visible faces of the second structure are hidden?



John 在桌上按照一定规律搭方块，从一个方块开始。接着，他在这个方块周围加上 5 个方块，将起始方块的各个可见面遮住，如图所示。现在，如果想继续将所有可见面遮住，John 至少还需要多少个方块？

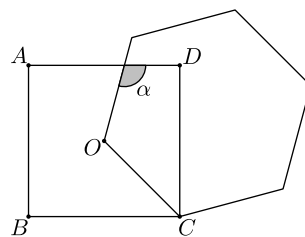
- (A) 8 (B) 9 (C) 10 (D) 13 (E) 19

8. A three-digit palindrome is a number of the form 'aba' where the digits a and b can either be the same or different. What is the sum of the digits of the largest three-digit palindrome that is also a multiple of 6?

三位回文数是指一系列形式为 'aba' 的数，其中数字 a 和数字 b 既可以相同也可以不同。请问是 6 的倍数的最大三位回文数的各位数字之和是多少？

- (A) 16 (B) 18 (C) 20 (D) 21 (E) 24

9. Martin draws a square with vertices A, B, C, D and a regular hexagon with side OC , where O is the center of the square. What is the size of angle α ?



Martin 画出一个顶点分别为 A, B, C, D 的正方形以及一个以 OC 为一条边的正六边形，其中 O 为正方形中心点。请问角 α 为多少度？

- (A) 105° (B) 110° (C) 115° (D) 120° (E) 125°

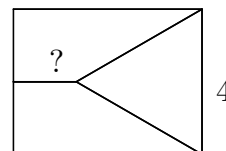
10. Ardal encloses a rectangular field with 40 m of fence. The side-lengths of the field are all prime numbers. What is the maximum possible area of the field?

Ardal 用 40 m 长的栅栏围住一个矩形区域，这个区域的边长都是质数。请问这个区域面积的最大可能值是多少？

- (A) 99 m^2 (B) 96 m^2 (C) 91 m^2 (D) 84 m^2 (E) 51 m^2

4 points

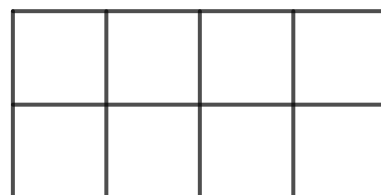
11. A rectangle is divided into three regions of equal area. One of the regions is an equilateral triangle with side-length 4 cm, the other two are trapezia, as shown in the figure. What is the length of the smaller of the parallel sides of the trapezia?



一个矩形被分为三个面积相同的区域。其中一个区域是边长为 4 cm 的等边三角形，其他两个区域为梯形，如图所示。请问梯形中较短平行边的长度是多少？

- (A) $\sqrt{2} \text{ cm}$ (B) $\sqrt{3} \text{ cm}$ (C) $2\sqrt{2} \text{ cm}$ (D) 3 cm (E) $2\sqrt{3} \text{ cm}$

12. Jelena places the capital letters A, B, C and D into the 2×4 table shown on the right. Exactly one letter is placed in each cell. She wishes to make sure that in each row and in each 2×2 square, each of the four letters appears exactly once. In how many ways can she do this?



右图为一个 2×4 的表格，Jelena 在其中放入大写字母 A、B、C、D。每个单元格中只放入一个字母。她希望这 4 个字母中的每个字母在每一行和每一个 2×2 的方格中只出现一次。请问 Jelena 一共有多少种放置方式？

- (A) 12 (B) 24 (C) 48 (D) 96 (E) 198

13. Sanjay cuts out three circles from three different pieces of coloured card. He places them on top of each other, as shown in Figure 1. He then moves the circles so that all three circles are tangent to each other, as shown in Figure 2. In the first figure, the area of the visible black region is seven times the area of the white circle. What is the ratio between the areas of the visible black regions in the two figures?

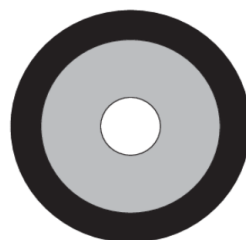


Figure 1
图 1

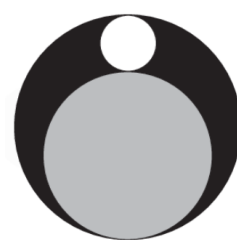


Figure 2
图 2

Sanjay 从三张不同的有色卡纸上剪下 3 张圆片，然后将 3 张圆片叠放在一起，如图 1 所示。紧接着 Sanjay 移动圆片，使得三张圆片两两相切，如图 2 所示。在第一张图中，黑色可见区域的面积是白色圆面积的 7 倍。请问两张图中黑色可见区域的面积之比是多少？

- (A) 3 : 1 (B) 4 : 3 (C) 6 : 5 (D) 7 : 6 (E) 9 : 7

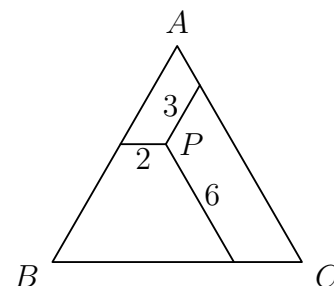
14. Mary's daughter gave birth to a baby girl today. In two years' time, the product of the ages of Mary, her daughter and her granddaughter will be 2024. Mary's and her daughter's ages are both even numbers. What is Mary's age now?

Mary 的孙女今天出生。两年后，Mary、Mary 的女儿和 Mary 的孙女，三人年龄的乘积等于 2024。已知 Mary 和 Mary 的女儿现在的年龄都是偶数，请问 Mary 现在多少岁？

- (A) 42 (B) 44 (C) 46 (D) 48 (E) 50

15. A point P is chosen inside an equilateral triangle. From P we draw three segments parallel to the sides, as shown. The lengths of the segments are 2 m, 3 m and 6 m. What is the perimeter of the triangle?

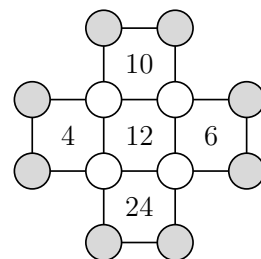
在等边三角形中选取点 P 。从点 P 出发，作三条线段分别与等边三角形的三条边平行，如图所示。这些线段的长度分别为 2 m、3 m、6 m。请问等边三角形的周长是多少？



- (A) 22 m (B) 26 m (C) 33 m (D) 39 m (E) 44 m

16. A number is written in each of the twelve circles shown. The number inside each square indicates the product of the numbers at its four vertices. What is the product of the numbers in the eight grey circles?

右侧图中有 12 个圆圈，每一个圆圈中都写有一个数字。每一个正方形内部的数字是这个正方形四个顶点圆圈中数字的乘积。请问 8 个灰色圆圈中数字的乘积是多少？



- (A) 20 (B) 40 (C) 80 (D) 120 (E) 480

17. There are four vases on the table in which a number of sweets have been placed.

The number of sweets in the first vase is the number of vases that contain one sweet.

The number of sweets in the second vase is equal to the number of vases that contain two sweets.

The number of sweets in the third vase is equal to the number of vases that contain three sweets.

The number of sweets in the fourth vase is equal to the number of vases that contain zero sweets.

How many sweets are in all the vases together?

桌上有 4 个花瓶，里面放了一些糖果。

第一个花瓶中糖果的数量等于包含 1 个糖果的花瓶的数量。

第二个花瓶中糖果的数量等于包含 2 个糖果的花瓶的数量。

第三个花瓶中糖果的数量等于包含 3 个糖果的花瓶的数量。

第四个花瓶中糖果的数量等于包含 0 个糖果的花瓶的数量。

请问所有花瓶中一共有多少个糖果？

- (A) 2 (B) 3 (C) 4 (D) 5 (E) 6

18. Jean-Philippe has n^3 ($n > 2$) identical small cubes. He used these to make a large cube and painted the entire outer surface of the large cube.

The number of small cubes with only one face painted is equal to the number of those with no face painted. What is the value of n ?

Jean-Philippe 有 n^3 ($n > 2$) 个相同的小立方体。他用这些小立方体做了一个大立方体，然后为这个大立方体的整个外表面上色。

其中只有一个面被上色的小立方体的数量等于完全未被上色的小立方体的数量。请问 n 的值是多少？

- (A) 4 (B) 6 (C) 7 (D) 8 (E) 10

19. Cristina has a set of cards numbered 1 to 12. She places eight of them at the vertices of an octagon so that the sum of every pair of numbers at opposite ends of an edge of the octagon is a multiple of 3.

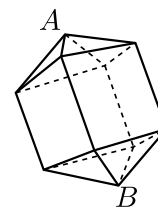
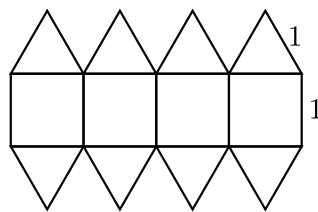
Which numbers did Cristina not place?

Cristina 有一套编号为 1 到 12 的卡片。她将其中八张卡片放在八边形的顶点上，使得八边形每条边两端的卡片编号之和为 3 的倍数。

请问 Cristina 没有将哪 4 张卡片放在八边形顶点上？

- (A) 1, 5, 9, 12 (B) 3, 5, 7, 9 (C) 1, 2, 11, 12 (D) 5, 6, 7, 8 (E) 3, 6, 9, 12

20. Otis makes a net using a combination of squares and equilateral triangles, as shown in the figure. The side-length of each square and of each triangle is 1 cm. He folds the net up into the 3D shape shown. What is the distance between the vertices A and B ?



Otis 通过组合正方形和等边三角形构成一个展开图，如图（左）所示。其中每个三角形和每个正方形的边长都是 1 cm。Otis 将该展开图折叠为一个 3D 图形，如图（右）所示。请问顶点 A 到顶点 B 的距离是多少？

- (A) $\sqrt{5}$ cm (B) $(1 + \sqrt{2})$ cm (C) $\frac{5}{2}$ cm
(D) $(1 + \sqrt{3})$ cm (E) $2\sqrt{2}$ cm

5 points

21. The prime factorisation of the number $n! = 1 \cdot 2 \cdot \dots \cdot n$ is of the form shown in the diagram.
数字 $n! = 1 \cdot 2 \cdot \dots \cdot n$ 的质因数分解如下：

$$2 \cdot 3 \cdot 5 \cdot 7 \cdot 11 \cdot 13^4 \cdot 17 \cdot \text{[Ink covered]} \cdot 43 \cdot 47$$

The primes are written in increasing order. Ink has covered some of the primes and some of the exponents. What is the exponent of 17?

所有质数按升序排列，其中一些质数和指数被墨水遮盖。请问 17 的指数是多少？

- (A) 1 (B) 2 (C) 3 (D) 4 (E) 5

22. Carl always tells the truth or always lies on alternate days. One day, he made exactly four of the following five statements. Which one could he not have made on that day?

Carl 总是一天说真话，一天说谎话，交替进行。某一天，Carl 恰好说了下列选项中的 4 句话。请问那天他并没有说下列哪一句话？

- (A) I lied yesterday and I will lie tomorrow.
我昨天说了谎话，明天也会说谎话。
(B) I am telling the truth today and I will tell the truth tomorrow.
今天我说的是真话，明天我也会说真话。
(C) 2024 is divisible by 11.
2024 可以被 11 整除。
(D) Yesterday was Wednesday.
昨天是星期三。
(E) Tomorrow will be Saturday.
明天是星期六。

23. The sum of the digits of the number N is three times the sum of the digits of the number $N + 1$. What is the smallest possible sum of the digits of N ?

数字 N 的各位数字之和是数字 $N + 1$ 的各位数字之和的三倍。请问数字 N 的各位数字之和的最小可能值是多少?

- (A) 9 (B) 12 (C) 15 (D) 18 (E) 27

24. Jill has some black, gray, and white unit cubes. She uses 27 of them to build a $3 \times 3 \times 3$ cube. She wants the surface to be exactly one-third black, one-third gray, and one-third white. The smallest possible number of black cubes she can use is A and the largest possible number of black cubes she can use is B . What is the value of $B - A$?

Jill 有一些黑色、灰色、白色的单位立方体。她用其中的 27 个立方体构建了一个 $3 \times 3 \times 3$ 的大立方体。她希望这个大立方体的表面上, $1/3$ 为黑色, $1/3$ 为灰色, $1/3$ 为白色。她最少用了 A 个黑色小立方体, 最多用了 B 个黑色小立方体。请问 $B - A$ 的值为多少?

- (A) 1 (B) 3 (C) 6 (D) 7 (E) 9

25. Ann rolled a normal die 24 times. All numbers from 1 to 6 came up at least once. The number 1 came up more times than any other number. Ann added up all the numbers. The total she obtained was the largest one possible. What total did she obtain?

Ann 将一枚常规骰子投掷了 24 次。从 1 到 6 的所有数字至少出现了一次。数字 1 出现的次数比其他任何数字出现的次数都多。Ann 将所有出现的数字相加, 得到的结果是最大可能值。请问她得到的结果是多少?

- (A) 83 (B) 84 (C) 89 (D) 90 (E) 100

26. Olya walked in the park. She walked half of the total time at a speed of 2 km/h. She walked half of the total distance at a speed of 3 km/h. She walked the rest of the time at a speed of 4 km/h. For what fraction of the total time did she walk at a speed of 4 km/h?

Olya 在公园里散步。她以 2 km/h 的速度走了总时长的一半, 以 3 km/h 的速度走了总路程的一半, 其余时间内的步行速度为 4 km/h。请问 Olya 以 4 km/h 的速度步行的时长占总时长的几分之几?

- (A) $\frac{1}{14}$ (B) $\frac{1}{12}$ (C) $\frac{1}{7}$ (D) $\frac{1}{5}$ (E) $\frac{1}{4}$

27. Ali wants to remove some of the integers from 1 to 25 and then separate the remaining numbers into two groups so that the products of the integers in each group are equal. What is the smallest number of integers Ali could remove?

Ali 想从 1 到 25 的所有整数中移除某些数字, 然后将剩下的数字分为两组, 使得每组整数的乘积相等。请问 Ali 最少可以移除多少个整数?

- (A) 4 (B) 5 (C) 6 (D) 7 (E) 8

28. Twenty points are equally spaced on the circumference of a circle. David draws all the possible chords that connect pairs of these points. How many of these chords are longer than the radius of the circle but shorter than its diameter?

在一个圆周上均匀分布着 20 个点。David 画出所有由这些点中任意两点构成的弦。请问其中有多少条弦的长度比圆的半径长且比直径短？

- (A) 90 (B) 100 (C) 120 (D) 140 (E) 160

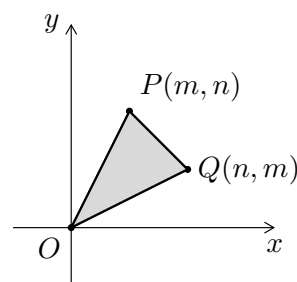
29. There are n distinct lines on the plane, labeled $\ell_1, \dots, \ell_n$. The line ℓ_1 intersects exactly 5 other lines, the line ℓ_2 intersects exactly 9 other lines, and the line ℓ_3 intersects exactly 11 other lines. Which of the following is a smallest possible value of n ?

平面上有 n 条不同的直线，标记为 $\ell_1, \dots, \ell_n$ 。直线 ℓ_1 恰好与另外 5 条直线相交，直线 ℓ_2 恰好与另外 9 条直线相交，直线 ℓ_3 恰好与另外 11 条直线相交。请问下列哪个选项为 n 的最小可能值？

- (A) 11 (B) 12 (C) 13 (D) 14 (E) 15

30. Suppose m and n are integers with $0 < m < n$. Let $P = (m, n)$, $Q = (n, m)$, and $O = (0, 0)$. For how many pairs of m and n will the area of triangle OPQ be equal to 2024?

假设 m 和 n 是整数，且 $0 < m < n$ 。设 $P = (m, n)$, $Q = (n, m)$, $O = (0, 0)$ ，请问存在多少对 m 和 n 可以使得三角形 OPQ 的面积等于 2024？



- (A) 4 (B) 6 (C) 8
(D) 10 (E) 12

Junior

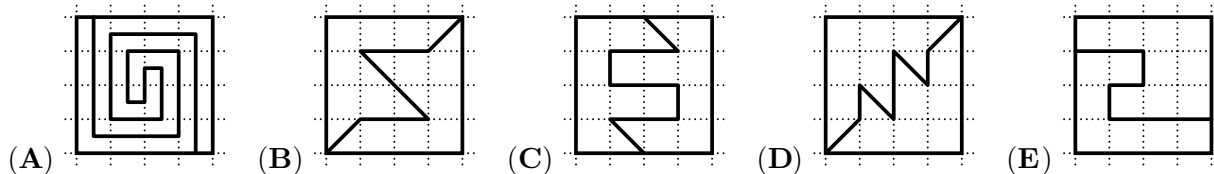
3 points

1 (Uganda). What is the value of $\frac{2 \times 0.24}{20 \times 2.4}$?

- (A) 0.01 (B) 0.1 (C) 1 (D) 10 (E) 100

SOLUTION: $\frac{2 \times 0.24}{20 \times 2.4} = \frac{0.48}{48} = 0.01$

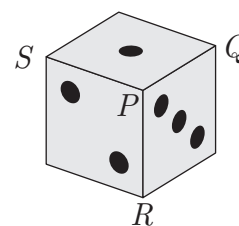
2 (Switzerland). Which square is split up into two pieces that do **not** have the same shape?



SOLUTION:

3 (Netherlands). The number of the dots on opposite faces of a die add to 7. The vertex labelled P on the die is formed by the faces which have 1, 2 and 3 dots on them. Its vertex sum is the sum of the number of dots on those faces which meet at a given vertex. The vertex sum of P is $1 + 2 + 3 = 6$. What is the maximum of the vertex sums of vertices Q , R and S ?

- (A) 7 (B) 9 (C) 10 (D) 11 (E) 15



SOLUTION: The vertex sum of Q is $1 + 3 + 5 = 9$, that of R is $2 + 3 + 6 = 11$ and that of S is $1 + 2 + 4 = 7$.

Hence the maximum is 11.

4 (Austria). A hopping game is played in the following way: Each player hops into the squares, swapping between left foot - both feet - right foot - both feet - left foot - both feet, and so on, as shown. Maya played the game and hopped into exactly 48 squares starting with her left foot. How many times did her left foot touch the ground?

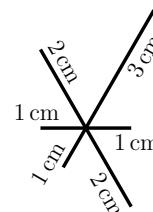
- (A) 12 (B) 24 (C) 36 (D) 40 (E) 48



SOLUTION: Maya's left foot touches the ground in three out of every four steps. Therefore, it touches the ground a total of $\frac{3}{4} \cdot 48 = 36$ times.

5 (Germany). Tim wants to draw the figure shown on a piece of paper, without lifting his pencil off the paper. The lengths of the lines are given in the figure. He can choose to start his drawing anywhere. What is the shortest distance he could draw to complete the figure?

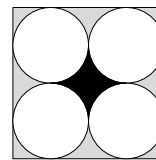
- (A) 14 cm (B) 15 cm (C) 16 cm
(D) 17 cm (E) 18 cm



SOLUTION: The shortest total distance is $3\text{cm} + 2 \cdot (1\text{cm} + 2\text{cm} + 1\text{cm} + 1\text{cm}) + 2\text{cm} = 15\text{cm}$.

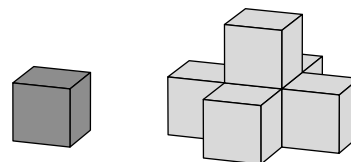
6 (Pakistan). The figure shows a square with four circles of equal area, each touching two sides of the square and two other circles. What is the ratio between the areas of the black region and the grey region?

- (A) 1 : 4 (B) 1 : 3 (C) 2 : 3 (D) 3 : 4 (E) $\pi : 1$



SOLUTION: The figure consists of 4 white circles inscribed into a quarter of a square. In each of those quarters, there is one black part and 3 parts of equally big grey area. So the answer is 1 : 3.

7 (Brazil). John makes a sequence of structures on a table, beginning with one cube. He makes the next structure by adding five cubes which hide the visible faces of the initial cube, as shown. What is the smallest number of cubes he needs to add to the second structure so that all the visible faces of the second structure are hidden?



- (A) 8 (B) 9 (C) 10 (D) 13 (E) 19

SOLUTION: John needs 1 cube on top, 4 cubes to cover 4 faces of the top cube and 8 cube on the ground level. It makes 13 cubes in total.

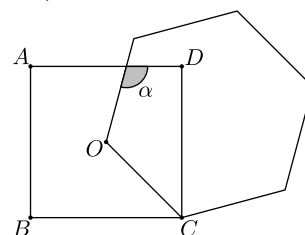
8 (Spain). A three-digit palindrome is a number of the form 'aba' where the digits a and b can either be the same or different. What is the sum of the digits of the largest three-digit palindrome that is also a multiple of 6?

- (A) 16 (B) 18 (C) 20 (D) 21 (E) 24

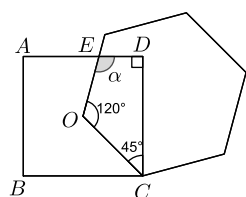
SOLUTION: If it is a multiple of 6, it must be even and its digits must add up to a multiple of three. It must start with 8 to also end in 8, the largest of which is 888 ($8 + 8 + 8 = 24$).

9 (Catalonia). Martin draws a square with vertices A, B, C, D and a regular hexagon with side OC , where O is the center of the square. What is the size of angle α ?

- (A) 105° (B) 110° (C) 115° (D) 120° (E) 125°



SOLUTION: Let E be the vertex of the requested angle. Look at the angles of the quadrilateral $OCDE$.



The angle in C is 45° (angle between side and diagonal of a square). The angle at D is a right angle. The angle at O is 120° (angle of the regular hexagon). Therefore $\alpha = 360^\circ - (45^\circ + 90^\circ + 120^\circ) = 105^\circ$.

10 (Myanmar). Ardal encloses a rectangular field with 40 m of fence. The side-lengths of the field are all prime numbers.

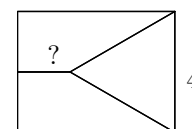
What is the maximum possible area of the field?

- (A) 99 m^2 (B) 96 m^2 (C) 91 m^2 (D) 84 m^2 (E) 51 m^2

SOLUTION: Perimeter of the fence $40 = 2 \cdot (l + w)$, so $l + w = 20$. Both l and w are prime numbers. There are 2 possible pairs $(l, w) = (3, 17), (7, 13)$. Maximum possible area is $7 \cdot 13 = 91 \text{ m}^2$

4 points

11 (Netherlands). A rectangle is divided into three regions of equal area. One of the regions is an equilateral triangle with side-length 4 cm, the other two are trapezia, as shown in the figure.

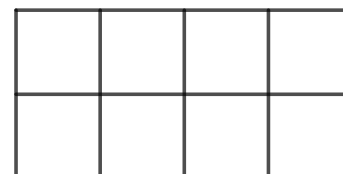


What is the length of the smaller of the parallel sides of the trapezia?

- (A) $\sqrt{2}$ cm (B) $\sqrt{3}$ cm (C) $2\sqrt{2}$ cm (D) 3 cm (E) $2\sqrt{3}$ cm

SOLUTION: The altitude of the triangle $2\sqrt{3}$, so its area is $\frac{1}{2} \cdot 4 \cdot 2\sqrt{3} = 4\sqrt{3}$. The area of the rectangle is then $3 \cdot 4\sqrt{3} = 12\sqrt{3}$. Then its length is $\frac{12\sqrt{3}}{4} = 3\sqrt{3}$. So the length of the smallest parallel side of the trapezia is $3\sqrt{3} - 2\sqrt{3} = \sqrt{3}$.

12 (Czech Republic). Jelena places the capital letters A, B, C and D into the 2×4 table shown on the right. Exactly one letter is placed in each cell. She wishes to make sure that in each row and in each 2×2 square, each of the four letters appears exactly once.



In how many ways can she do this?

- (A) 12 (B) 24 (C) 48 (D) 96 (E) 198

SOLUTION: The letters A, B, C and D can be arranged freely in the four squares of the first row. There are four options for the first letter, then three options for the second square and two for the third square. That is $4 \times 3 \times 2 \times 1$, which is equal to 24 options.

Having filled the first row, consider the second square of the second row. This can't be the letter in the first or third position of the first row. So it must be the same letter as in the fourth position. Therefore the second row is uniquely defined by the first row and there are 24 options.

13 (Brazil). Sanjay cuts out three circles from three different pieces of coloured card. He places them on top of each other, as shown in Figure 1. He then moves the circles so that all three circles are tangent to each other, as shown in Figure 2.

In the first figure, the area of the visible black region is seven times the area of the white circle.

What is the ratio between the areas of the visible black regions in the two figures?

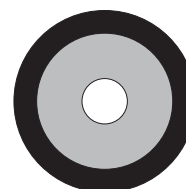


Fig 1

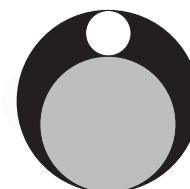


Fig 2

- (A) 3 : 1 (B) 4 : 3 (C) 6 : 5 (D) 7 : 6 (E) 9 : 7

SOLUTION: Let the area of white, black and grey circles respectively be w , b , g . Also if the visible black area in figure 1 are $A_1 = b - g = 7w$ and in figure 2, $A_2 = b - w - g$ so $A_2 = 7w - w = 6w$ (note that $w < g$).

14 (Hungary). Mary's daughter gave birth to a baby girl today. In two years' time, the product of the ages of Mary, her daughter and her granddaughter will be 2024.

Mary's and her daughter's ages are both even numbers. What is Mary's age now?

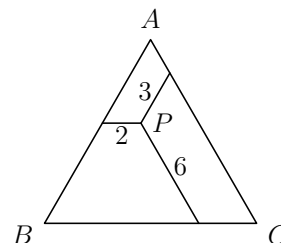
- (A) 42 (B) 44 (C) 46 (D) 48 (E) 50

SOLUTION: The granddaughter will be two years old in two years time. Let Mary have the age of $2m$ and her daughter's age be $2d$. The product of the ages must be $2 \cdot (2d) \cdot (2m)$. We compare this with the prime factorisation of $2024 = 2 \times (2 \cdot 11) \times (2 \cdot 23)$.

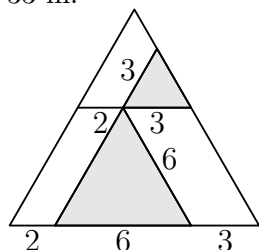
So Mary will be 46 in two years time, making her age now 44.

15 (Greece). A point P is chosen inside an equilateral triangle. From P we draw three segments parallel to the sides, as shown. The lengths of the segments are 2 m, 3 m and 6 m. What is the perimeter of the triangle?

- (A) 22 m (B) 26 m (C) 33 m (D) 39 m (E) 44 m

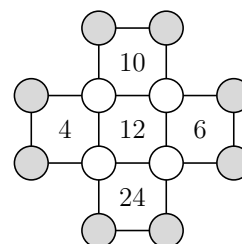


SOLUTION: Extend one segment through P . We have an equilateral triangle and parallelogram. The sides of the original triangle have length $2\text{ m} + 3\text{ m} + 6\text{ m} = 11\text{ m}$. So the perimeter of the triangle is 33 m.

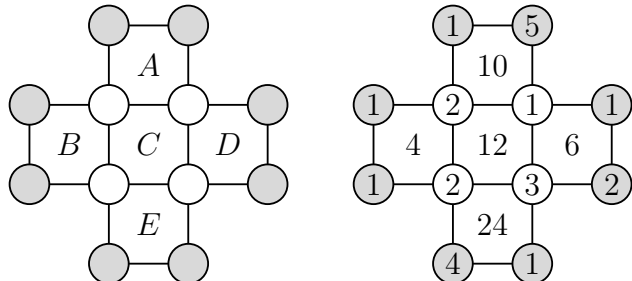


16 (Greece). A number is written in each of the twelve circles shown. The number inside each square indicates the product of the numbers at its four vertices. What is the product of the numbers in the eight grey circles?

- (A) 20 (B) 40 (C) 80
(D) 120 (E) 480



SOLUTION: If the products inside the squares are A, B, C, D, E , then the required product is $ABDE/C^2$. This is because the numbers in grey circles are included once each at the products A, B, D and E and those in white circles show up twice each. The picture on the right shows an example with a possible construction.



17 (Latvia). There are four vases on the table in which a number of sweets have been placed. The number of sweets in the first vase is the number of vases that contain one sweet. The number of sweets in the second vase is equal to the number of vases that contain two sweets. The number of sweets in the third vase is equal to the number of vases that contain three sweets. The number of sweets in the fourth vase is equal to the number of vases that contain zero sweets. How many sweets are in all the vases together?

- (A) 2 (B) 3 (C) 4 (D) 5 (E) 6

SOLUTION: If in any of the 4 vases would be more than 4 sweets that would lead to a contradiction because there are only 4 vases. If there were 4 sweets in one of the 4 vases, it would lead to a contradiction because this vase itself contains a different number of sweets. So there are only vases with 0, 1, 2 or 3 sweets. The total number of sweets is equal to the number of vases. There are at least two possible distributions in four vases: 2, 1, 0, 1 or 0, 2, 0, 2.

18 (Catalonia). Jean-Philippe has $n^3 (n > 2)$ identical small cubes. He used these to make a large cube and painted the entire outer surface of the large cube.

The number of small cubes with only one face painted is equal to the number of those with no face

Painted. What is the value of n ?

(A) 4

(B) 6

(C) 7

(D) 8

(E) 10

SOLUTION: There are $(n-2)^2$ small cubes with a single painted face visible on each face of the large cube; so the total number is $6(n-2)^2$. There are $(n-2)^3$ cubes without any outer face and the big cube. From $6(n-2)^2 = (n-2)^3$ and $n > 2$ we deduce $n = 8$.

19 (Greece). Cristina has a set of cards numbered 1 to 12. She places eight of them at the vertices of an octagon so that the sum of every pair of numbers at opposite ends of an edge of the octagon is a multiple of 3.

Which numbers did Cristina not place?

(A) 1, 5, 9, 12

(B) 3, 5, 7, 9

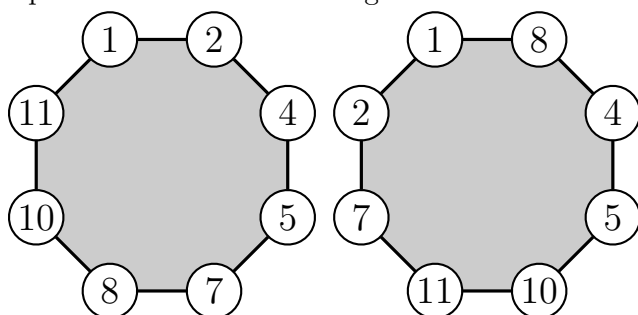
(C) 1, 2, 11, 12

(D) 5, 6, 7, 8

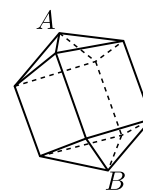
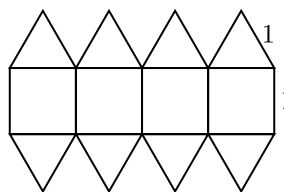
(E) 3, 6, 9, 12

SOLUTION: If one of the numbers at a vertex is a multiple of 3 then the number at EVERY neighbouring vertex must also be a multiple of 3. Since we have only 4 multiples of 3 we come to a contradiction. So the numbers 3, 6, 9, 12 must be removed.

2 possible constructions are given below.



20 (Mexico). Otis makes a net using a combination of squares and equilateral triangles, as show in the figure. The side-length of each square and of each triangle is 1 cm. He folds the net up into the 3D shape shown. What is the distance between the vertices A and B ?



- (A) $\sqrt{5}$ cm (B) $(1 + \sqrt{2})$ cm (C) $\frac{5}{2}$ cm
(D) $(1 + \sqrt{3})$ cm (E) $2\sqrt{2}$ cm

SOLUTION: The 4 vertices joined to A form a square $CDEF$ of side 1 cm long. Therefore, by the Pythagorean theorem, its diagonal CE is $\sqrt{2}$ cm long. Let M be the midpoint of CE . Then AMC is a right triangle with hypotenuse AC of side 1 cm and leg CM of side $\frac{\sqrt{2}}{2}$ cm. Again, using the Pythagorean theorem, we get

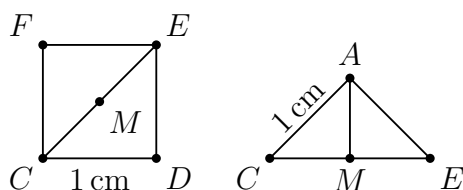
$$|AM|^2 = 1 - \left(\frac{\sqrt{2}}{2}\right)^2 = 1 - \frac{2}{4} = \frac{1}{2},$$

thus

$$|AM| = \frac{1}{\sqrt{2}} = \frac{\sqrt{2}}{2}.$$

The distance between A and B is

$$\frac{\sqrt{2}}{2} + 1 + \frac{\sqrt{2}}{2} = 1 + \sqrt{2}.$$



5 points

21 (Greece). The prime factorisation of the number $n! = 1 \cdot 2 \cdot \dots \cdot n$ is of the form shown in the diagram.

$$2 \cdot 3 \cdot 5 \cdot 7 \cdot 11 \cdot 13^4 \cdot 17 \cdot \text{[Ink]} \cdot 43 \cdot 47$$

The primes are written in increasing order. Ink has covered some of the primes and some of the exponents. What is the exponent of 17?

- (A) 1 (B) 2 (C) 3 (D) 4 (E) 5

SOLUTION: The appearance of the prime 47 shows that $47 \leq n < 53$ (else we would see the next prime 53). Since 13^4 is part of the factorization, 4 multiples of 13 are included in $n!$. These are 13, 26, 39, and 52, so $n = 52$. There are three multiples of 17 before 52 (these are 17, 34 and 51). Therefore, the exponent of 17 in this factorization is 3.

22 (Hungary). Carl always tells the truth or always lies on alternate days. One day, he made exactly four of the following five statements. Which one could he not have made on that day?

- (A) I lied yesterday and I will lie tomorrow.
(B) I am telling the truth today and I will tell the truth tomorrow.
(C) 2024 is divisible by 11.

(D) Yesterday was Wednesday.

(E) Tomorrow will be Saturday.

SOLUTION: Statement (B) can only be made by a person who lies on that day. Looking at the statements (D) and (E), they cannot both be true. Therefore, we have at least two statements that are lies. This means that Carl is lying today, so he couldn't make the statement (C).

23 (Russia). The sum of the digits of the number N is three times the sum of the digits of the number $N + 1$. What is the smallest possible sum of the digits of N ?

(A) 9

(B) 12

(C) 15

(D) 18

(E) 27

SOLUTION: The sum of the digits of a number decreases only if the number has the last digit 9, so that the next number ends in 0, and 1 is added to its penultimate digit. In that case, the sum of the digits of the number $N + 1$ decreases by 8, compared to that of the number N . At the same time, it is three times smaller than the sum of the digits of N . Denoting the sum of the digits of N by x , we have

$$x = 3(x - 8) \Leftrightarrow 2x = 24 \Leftrightarrow x = 12.$$

Note that if the number ends in more than one 9, its sum of digits will be at least 19, which is already greater than 12. The number 39 has the property described above.

24 (Austria). Jill has some black, gray, and white unit cubes. She uses 27 of them to build a $3 \times 3 \times 3$ cube. She wants the surface to be exactly one-third black, one-third gray, and one-third white. The smallest possible number of black cubes she can use is A and the largest possible number of black cubes she can use is B . What is the value of $B - A$?

(A) 1

(B) 3

(C) 6

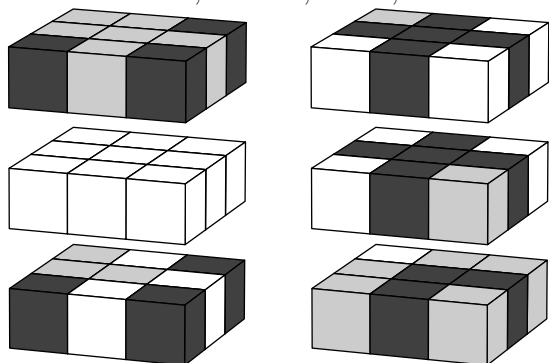
(D) 7

(E) 9

SOLUTION: The surface area of the cube is $6 \cdot 3^2 = 54$. This means that the black, the grey and the white area should equal to 18 each.

Each unit cube can contribute 3 if it is a corner cube, 2 if it is an edge cube, and 1 if it is a center of a face.

There are 8 corner cubes, 12 edge cubes and 6 centers of faces. 1 unit cube is completely hidden from the surface. The smallest number of unit cubes to make area of 18 is $A = 6$ corner cubes. The largest number of unit cubes to make area of 18 is 12: 6 centers of faces and 6 edge cubes. Indeed, we can choose the "hidden" unit cube to have the same color, so we would have used $B = 13$ black unit cubes. In other words, $B = 13$, $A = 6$, $B - A = 7$.



25 (Catalonia). Ann rolled a normal die 24 times. All numbers from 1 to 6 came up at least once. The number 1 came up more times than any other number. Ann added up all the numbers. The total she obtained was the largest one possible. What total did she obtain?

(A) 83

(B) 84

(C) 89

(D) 90

(E) 100

SOLUTION: Since Ann has rolled each number at least once, we can guarantee that 6 of the rolls have a sum of $1 + 2 + 3 + 4 + 5 + 6 = 21$. Let's consider the remaining 18 rolls.

There should be more 1s than any other numbers. It is clear that the number of 6s rolled should be one less than the number of 1s.

If we have 9 1s: then 8 6s adding a 5 we get $9 + 48 + 5 = 62$.

If we have 8 1s: then 7 6s adding 3 5s we get $8 + 42 + 15 = 65$.

If we have 7 1s: then 6 6s adding 5 5s we get $7 + 36 + 25 = 68$.

If we have 6 1s: then 5 6s, 5 5s and adding 2 4s we get $6 + 30 + 25 + 8 = 69$. This leads to a max total of 90.

Further reduction of 1s does not increase the maximum possible sum.

26 (Russia). Olya walked in the park. She walked half of the total time at a speed of 2 km/h. She walked half of the total distance at a speed of 3 km/h. She walked the rest of the time at a speed of 4 km/h. For what fraction of the total time did she walk at a speed of 4 km/h?

- (A) $\frac{1}{14}$ (B) $\frac{1}{12}$ (C) $\frac{1}{7}$ (D) $\frac{1}{5}$ (E) $\frac{1}{4}$

SOLUTION: Let's denote the half of the time for Olya's walk with t and the time she walks with a speed of 4 km/h with x . Then, the time she walks with a speed of 3 km/h is $t - x$. We have the following:

Speed: 2 km/h, Time: t , Distance : $2t$

Speed: 3 km/h, Time: $t - x$, Distance : $3(t - x)$

Speed: 4 km/h, Time: x , Distance : $4x$

Then we have

$$3(t - x) = 2t + 4x \Leftrightarrow t = 7x \Leftrightarrow x = t/7$$

Therefore, $x = 1/14(2t)$. That is, $1/14$ of the time was spent walking with a speed of 4 km/h.

27 (Turkey). Ali wants to remove some of the integers from 1 to 25 and then separate the remaining numbers into two groups so that the products of the integers in each group are equal. What is the smallest number of integers Ali could remove?

- (A) 4 (B) 5 (C) 6 (D) 7 (E) 8

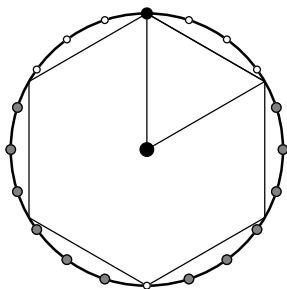
SOLUTION: Thinking of the prime factorizations of the products of the numbers of both groups, they have to be the same, since the products are equal. Therefore, prime numbers that have only one multiple among the numbers from 1 to 25 have to be removed. These prime numbers are 13, 17, 19 and 23. 7 has three multiples among the numbers from 1 to 25: 7, 14 and 21. At least one of them must be removed, which works only with the deletion of 7. The remaining 20 numbers can be split in two groups that have equal product.

For example:

$$3 \cdot 5 \cdot 8 \cdot 14 \cdot 15 \cdot 18 \cdot 20 \cdot 22 \cdot 24 = 2 \cdot 4 \cdot 6 \cdot 9 \cdot 10 \cdot 11 \cdot 12 \cdot 16 \cdot 21 \cdot 25 = 2^{11} \cdot 3^5 \cdot 5^3 \cdot 7 \cdot 11.$$

28 (Hong Kong). Twenty points are equally spaced on the circumference of a circle. David draws all the possible chords that connect pairs of these points. How many of these chords are longer than the radius of the circle but shorter than its diameter?

- (A) 90 (B) 100 (C) 120 (D) 140 (E) 160



SOLUTION:

Each of the 20 points is connected with the 19 other points on the circumference, forming 19 chords. One of these 19 chords is exactly equal to the diameter, leaving 18 chords to consider.

For a chord to be larger than the radius, its central angle has to be larger than 60° .

Since the 20 points form a regular 20-gon, the central angle of the chord connecting two adjacent points is $360/20 = 18^\circ$. Therefore, we need at least 4 central angles ($3 \cdot 18^\circ = 54^\circ < 60^\circ$).

That is, 3 pairs of chords from a point to its three nearest points along the circumference (on either side) are going to be shorter than the radius. This leaves us with 6 pairs of chords, or 12 chords per point. Since each chord is counted twice (once per endpoint) the total number of chords satisfying the condition is $(20 \cdot 12) : 2 = 120$.

29 (Belarus). There are n distinct lines on the plane, labeled $\ell_1, \dots, \ell_n$. The line ℓ_1 intersects exactly 5 other lines, the line ℓ_2 intersects exactly 9 other lines, and the line ℓ_3 intersects exactly 11 other lines. Which of the following is a smallest possible value of n ?

(A) 11

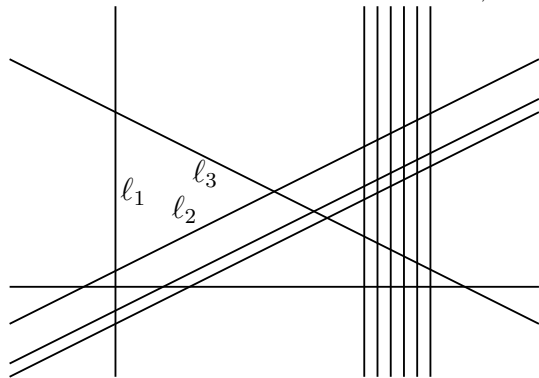
(B) 12

(C) 13

(D) 14

(E) 15

SOLUTION: Since one of the lines intersects 11 other lines, the minimum number of lines is 12. On the other hand, taking a family of 7 parallel lines, another family of 3 parallel lines perpendicular to the first family and two other lines, intersecting each other and intersecting both families of parallel lines, we have an arrangement of lines that conforms to the requirement. Any of the lines in the first family of 7 parallel lines can be ℓ_1 , as it intersects 5 other lines. Any of the lines in the second family of 3 parallel lines can be ℓ_2 as it intersects 9 other lines. Finally, any of the last two lines can be ℓ_3 as it intersects 11 other lines. Therefore, the answer is 12.



30 (China). Suppose m and n are integers with $0 < m < n$. Let $P = (m, n)$, $Q = (n, m)$, and $O = (0, 0)$. For how many pairs of m and n will the area of triangle OPQ be equal to 2024?

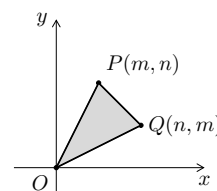
(A) 4

(B) 6

(C) 8

(D) 10

(E) 12



SOLUTION: $A_{\triangle OPQ} = n^2 - mn - \frac{1}{2}(n-m)^2 = \frac{1}{2}(n^2 - m^2) = 2024 \implies (n+m)(n-m) = 4048$.

Since m and n are both integers $n+m$ and $n-m$ are both even or both odd.

There are six ways that $4048 = 2^4 \cdot 11 \cdot 23$ can be represented as a product of two even numbers: 2×2024 , $2^2 \times 1012$, $2^3 \times 506$, $(2 \cdot 11) \times (2^3 \cdot 23)$, $(2^2 \cdot 11) \times (2^2 \cdot 23)$, $(2 \cdot 23) \times (2^3 \cdot 11)$, so there are 6 possible pairs of $(n, m) : \{(1013, 1011), (508, 504), (257, 249), (103, 81), (68, 24), (67, 21)\}$.